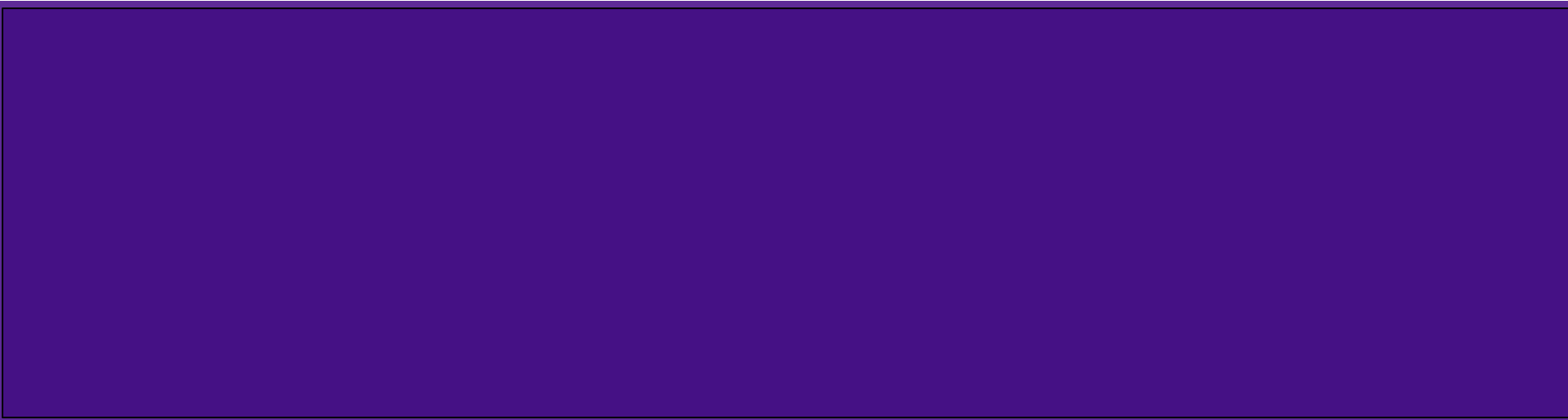




# CSDE 502

## Algorithms & Prediction

May 9, 2025



# Research, Prediction, & Racism (Roberts & Rollins, 2020)

## Measurements of Racial Difference

- **Genetics and Biological Determinism**
  - Eugenics
  - Post-genomic era & the “molecularization of race”
- **Social Construction**
  - Racism and racial inequity
  - Racial identity & belonging
- **Where they clash**
  - Ancestry, “marketization of identity”
  - DNA testing, citizenship, & family separation
- **Biosocial**
  - DuBois & structural racism
  - Epigenetics, allostatic load
  - Environmental racism, segregation, state violence
- **Biosocial (Neuroscience)**
  - “Fixing racism” for White people
  - Criminology
  - Education



# Algorithmic Fairness (Mitchell et al, 2021)



## Assumptions

- What are we trying to predict and why?
- Population and population mismatches
- Possible decisions

## Bias

- Societal bias
- Sampling bias
- Measurement Error

## Evaluation

- Accuracy
- Fairness

## Fairness

- Single-threshold, i.e. all individuals with same covariates get equal decision
- Same accuracy across subgroups
- Same decisions across subgroups, conditional on outcome
  - same rate of true +/false - & true - /false +
- Same outcomes across subgroups, conditional on decision
- Same predictions across subgroups, conditional on outcome
- Same fairness metric value based on joint distribution of covariates, subgroup, decision, outcome